



Fig. 4: Protocol for mobile IP

the home agent, the home agent works as a proxy of the mobile node. When the home network receives packets addressed to the mobile node, the home agent intercepts those packets (using proxy ARP) and encapsulates them. The encapsulation is done with the overhead care-of address. The encapsulated packets are then sent to the care-of address—the current IP address of the foreign agent with which the mobile node is registered. The foreign agent decapsulates those packets, and forwards them to the mobile node (because it knows exactly where the mobile node is).

Encapsulation is the method used by the home agent to deliver information to the mobile node by putting an extra IP header on top of the packet. Tunneling is a process by which the encapsulated packet is sent to the mobile node when it is in the foreign network. Two forms of encapsulation are specified in related standards: Request For Comments/RFC 2003 (IP encapsulation within IP) and 2004 (minimal encapsulation within IP).

Getting recognition (discovery) makes use of the Internet control message protocol (ICMP). This is done by appending appropriate extension to the ICMP headers. ICMP is a connectionless protocol.

Registration requires a protocol for communication between the mobile node and home agent applications. Thus it requires a transport-level protocol. As it is a simple request-and-response transaction, the overhead of the connection-oriented TCP is not required. Therefore user datagram

protocol (UDP) is used as the transport protocol for registration. Tunneling is done at the IP level. Mobile IP protocol is shown in Fig. 4.

Mobile IP with IPv6

As said earlier, mobile IP is meant to provide mobility support on

the existing IP infrastructure, without requiring any modification to the existing routers, applications or fixed-end hosts. So how does it change with IP version 6?

Mobile IPv6 allows an IPv6 node to be mobile, just like an IPv4 node. As opposed to mobile IPv4, it is intended to provide additional supports for mobility, route optimisation and security. It will allow the Internet nodes to associate a mobile node's home address with its care-of address and send packets directly to the care-of address rather than via the home address. This will eliminate any triangular routing.

In fact, IPv6 improvement comes from two associated standards, Neighbor Discovery for IP Version 6 (RFC 1970) and IPv6 Stateless Address Autoconfiguration (RFC 1971). By this a mobile node obtains a care-of address, eliminating the requirement for foreign agents. Mobile IPv6 uses route optimisation, by which the mobile node sends its current care-of address to the correspondent node using binding update. When it knows the mobile node's current care-of address, a correspondent node can deliver packets directly to the mobile node's home address without any assistance from the home agent.

Unlike IPv4, mobile IPv6 does not make use of foreign agent but correspondent node and binding. Binding refers to an association of the home address with care-of address of the mobile node. The correspondent node is a peer node with which a mobile node communicates. It may either be

mobile or stationary.

Mobile IPv6 operates as follows:

1. The mobile node gets registration to the foreign network and gets care-of address as in mobile IPv4
2. The mobile node sends binding update to the home agent
3. The home agent uses proxy neighbour discovery to find mobile node in the home network
4. The IP datagram addressed to the mobile node is encapsulated in an IPv6-to-IPv6 tunnel. Then the encapsulated packet is sent to care-of address of the mobile node
5. The IP datagram from the mobile node makes use of the same tunnel in reverse means
6. The IP datagram from the mobile node is sent to the correspondent node directly
7. The mobile node sends binding update to the correspondent node

Route optimisation is likely to dramatically improve the performance of IPv6 mobile nodes. Another improvement is the development of the standard IP authentication header (RFC 1826). Mobility support in IPv6 retains the ideas of a home network, home agent and encapsulation to deliver packets from the home network to the mobile node's current point of attachment, as in IPv4. Foreign agents are not required to support mobility in IPv6. IPv6-within-IPv6 tunneling is also already specified. IPv6 will improve routing efficiency, enhance the mobility support and relatively simplify procedures.

Challenges ahead

There are still some technical and developmental obstacles which need to be addressed in order to realise fully mobile networking over widespread coverage. The most critical issue is how to integrate mobile IP with existing IP network (non-mobile, wired or wireless). Inter-working with the fixed network on a wide scale remains a big challenge. ●

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